MAT 216
Lec 10 (Week 5)
Highlights

* You must see the Video Lec 10 in buX
* You must read the Lec NOTE from buX.

Column Space:

$$A = \begin{array}{c} \begin{array}{cc} v_1 & v_2 \end{array} \\ \begin{bmatrix} a_1 & b_1 \\ a_2 & b_2 \\ a_3 & b_3 \end{bmatrix} \end{array} \begin{array}{c} \\ 1 \\ 2 \\ 3 \end{array}$$

3 elements

$$C(A) = c_1 v_1 + c_2 v_2$$

$$C(A) = \text{Span}(v_1, v_2) \longrightarrow \text{Subspace in } \mathbb{R}^3$$

The Column of a matrix that span the column Space might be dependent or independent

(i) $\begin{bmatrix} 1 & 2 & 5 \\ 0 & 3 & 9 \end{bmatrix}$ no. of col. vectors > no. of dimension of vector space

$\downarrow$ 3 $\qquad$ $\downarrow$ 2

The vectors must be dependent

Any 3 vectors in $\mathbb{R}^2$ must be dependent.

(ii) $\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$ no. of col. vectors $\leq$ no. of dimension of vector space

The vectors might be dependent / independent.

$\begin{bmatrix} 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ 4 \end{bmatrix} \longrightarrow$ These 2 vectors are dependent. So we are getting a subspace. a line in $\mathbb{R}^2$.

$\begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \longrightarrow$ In this case we can construct the entire 2-dimension ($\mathbb{R}^2$)

(xy-plane)

Row Space:

$$\begin{bmatrix} a_1 & b_1 \\ a_2 & b_2 \\ a_3 & b_3 \end{bmatrix}$$

The row vectors are $v_1 = [a_1 \quad b_1]$

$v_2 = [a_2 \quad b_2]$

$v_3 = [a_3 \quad b_3]$

$R(A) = c_1 v_1 + c_2 v_2 + c_3 v_3 = \text{Span}(v_1, v_2, v_3)$

Thus $R(A)$ is a subspace ~~of~~ in $\mathbb{R}^2$

Note: $R(A) = C(A^T)$

$$A = \begin{bmatrix} a_1 & b_1 \\ a_2 & b_2 \\ a_3 & b_3 \end{bmatrix} \longrightarrow A^T = \begin{bmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{bmatrix}$$

(i) & (ii)

Row Space & Column Space

Theorem If a matrix R is in row echelon form, then the row vectors with pivots forms the basis for the row space of R, and the column vectors with pivots of the row vectors form a basis for the column space of R.

Let $A = \begin{bmatrix} 1 & 3 & 2 \\ 2 & 6 & 4 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix} \xrightarrow[\text{Reduction}]{\text{Row}} \begin{bmatrix} \boxed{1} & 0 & -1 \\ 0 & \boxed{1} & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ = lets call this matrix R

Pivot

The Basis of Row space:

$$R(A) = \{(1,0,-1), (0,1,1)\}$$

OR $\left\{ \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \right\}$

The Basis of Column space:

$$C(A) = \left\{ \begin{bmatrix} 1 \\ 2 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 3 \\ 6 \\ 1 \\ 0 \end{bmatrix} \right\}$$

$\because$ after row reduction the 1st and 2nd columns contain the pivots in R hence the corresponding columns of matrix A forms the basis of column space.

Rank (A) = 2 $\because$ There are 2 independent vectors.

The dimension of a vector space is equal to the maximum number of linearly independent vectors contained in it. Hence Rank(A) = dimension of C(A) = dimension of R(A)

(iii) Null Space or Solution space

The solution space of the homogeneous system of equations $Ax = 0$, which is a subspace of $\mathbb{R}^n$, is called the nullspace of A.

Ex (a) $A = \begin{bmatrix} 1 & 3 & 2 \\ 2 & 6 & 4 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix} \xrightarrow[\text{by row echelon form}]{\text{reduced}} \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

Now consider $Ax = 0$ Form

$$\begin{bmatrix} \boxed{1} & 0 & -1 \\ 0 & \boxed{1} & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

pivots

x_3 is the free variable

let $\boxed{x_3 = p}$ any arbitrary value

$R_2 : x_2 + x_3 = 0$

$x_2 = -x_3$

$\boxed{x_2 = -p}$

$R_1 : x_1 - x_3 = 0$

$x_1 = x_3$

$\boxed{x_1 = p}$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} p \\ -p \\ p \end{bmatrix} = p \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$$

Nullity = 1 [total number of freevariables]

Null space : $N(A) = \left\{ \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} \right\}$

Ex(b) Find the solution space of

$$A = \begin{bmatrix} -1 & 2 & 0 & 4 & 5 & -3 \\ 3 & -7 & 2 & 0 & 1 & 4 \\ 2 & -5 & 2 & 4 & 6 & 1 \\ 4 & -9 & 2 & -4 & -4 & 7 \end{bmatrix}$$

$\xrightarrow{\text{Reduced by row echelon form}}$ $\begin{bmatrix} 1 & 0 & -4 & -28 & -37 & 13 \\ 0 & 1 & -2 & -12 & -16 & 5 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

Consider $Ax = 0$ form

$$\begin{bmatrix} \boxed{1} & 0 & -4 & -28 & -37 & 13 \\ 0 & \boxed{1} & -2 & -12 & -16 & 5 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \\ x_6 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

pivots/leading 1

x_3, x_4, x_5, x_6 are free variables

Let $\boxed{x_3 = r}$, $\boxed{x_4 = s}$, $\boxed{x_5 = t}$, $\boxed{x_6 = u}$ any arbitrary value.

R_2: $x_2 = 2x_3 + 12x_4 + 16x_5 - 5x_6$

$$\boxed{x_2 = 2r + 12s + 16t - 5u}$$

R_1: $x_1 = 4x_3 + 28x_4 + 37x_5 - 13x_6$

$$\boxed{x_1 = 4r + 28s + 37t - 13u}$$

$$\therefore \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \\ x_6 \end{bmatrix} = \begin{bmatrix} 4r + 28s + 37t - 13u \\ 2r + 12s + 16t - 5u \\ r \\ s \\ t \\ u \end{bmatrix} = r\begin{bmatrix} 4 \\ 2 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + s\begin{bmatrix} 28 \\ 12 \\ 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} + t\begin{bmatrix} 37 \\ 16 \\ 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} + u\begin{bmatrix} -13 \\ -5 \\ 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

Nullity $= 4$

Nullspace: $N(A) = \left\{ \begin{bmatrix} 4 \\ 2 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 28 \\ 12 \\ 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 37 \\ 16 \\ 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -13 \\ -5 \\ 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} \right\}$